Band-1 Hadamard order 13968 – Classical Paley-I (q=13967); IN SPK via Paley-I; Gap 2052 Still OPEN

H_13968 Paley-I PASS; Paley-II FAIL/SKIP (q=6983 prime but ===3 mod 4 needs ===1); Kronecker SKIP deferred note-only (*43492; 3492===0 mod 4 order-admissible; factors 2^23^2*97*; not_in_seeds / no H_3492; H_3492 classically via Paley-I q=3491 prime ===3 mod 4 (direct) ALSO ELIGIBLE – deferred note-only, NOT constructed, not on box); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-10-07

Document class. Scientific computational / construction certificate (Band-1).

Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.

Parents. `2026-10-07_Band1_Hadamard_13964_PaleyI`; `2026-10-07_Band1_Hadamard_13932_PaleyI`; `2026-10-07_Band1_Hadamard_13908_PaleyI`; `2026-10-07_Band1_Hadamard_13904_PaleyI`; `2026-10-07_Band1_Hadamard_13884_PaleyI`; `2026-10-07_Band1_Hadamard_13880_PaleyI`; `2026-10-07_Band1_Hadamard_13860_PaleyI`; `2026-10-07_Band1_Hadamard_13832_PaleyI`; `2026-10-07_Band1_Hadamard_13808_PaleyI`; `2026-10-07_Band1_Hadamard_13800_PaleyI`; `2026-10-07_Band1_Hadamard_13764_PaleyI`; `2026-10-07_Band1_Hadamard_13760_PaleyI`; `2026-10-07_Band1_Hadamard_13752_PaleyI`; `2026-10-07_Band1_Hadamard_13724_PaleyI`; `2026-10-07_Band1_Hadamard_13712_PaleyI`; `2026-10-07_Band1_Hadamard_13692_PaleyI`; `2026-10-07_Band1_Hadamard_13688_PaleyI`; `2026-10-06_Band1_Hadamard_13680_PaleyI`; `2026-10-06_Band1_Hadamard_13628_PaleyI`; `2026-10-06_Band1_Hadamard_13620_PaleyI`; `2026-10-06_Band1_Hadamard_13592_PaleyI`; `2026-10-06_Band1_Hadamard_13568_PaleyI`; `2026-10-06_Band1_Hadamard_13524_PaleyI`; `2026-10-06_Band1_Hadamard_13500_PaleyI`; `2026-10-06_Band1_Hadamard_13488_PaleyI`; `2026-10-06_Band1_Hadamard_13464_PaleyI`; `2026-10-06_Band1_Hadamard_13452_PaleyI`; `2026-10-06_Band1_Hadamard_13412_PaleyI`; `2026-10-06_Band1_Hadamard_13400_PaleyI`; `2026-10-06_Band1_Hadamard_13368_PaleyI`; `2026-10-06_Band1_Hadamard_13340_PaleyI`; `2026-10-06_Band1_Hadamard_13332_PaleyI`; `2026-10-06_Band1_Hadamard_13328_PaleyI`; `2026-10-06_Band1_Hadamard_13292_PaleyI`; `2026-10-06_Band1_Hadamard_13268_PaleyI`; `2026-10-06_Band1_Hadamard_13260_PaleyI`; `2026-10-06_Band1_Hadamard_13220_PaleyI`; `2026-10-06_Band1_Hadamard_13188_PaleyI`; `2026-10-06_Band1_Hadamard_13184_PaleyI`; `2026-10-06_Band1_Hadamard_13172_PaleyI`; `2026-10-05_Band1_Hadamard_13164_PaleyI`; `2026-10-05_Band1_Hadamard_13160_PaleyI`; `2026-10-05_Band1_Hadamard_13152_PaleyI`; `2026-10-05_Band1_Hadamard_13148_PaleyI`; `2026-10-05_Band1_Hadamard_13128_PaleyI`; `2026-10-05_Band1_Hadamard_13104_PaleyI`;

2026-10-05_Band1_Hadamard_13100_PaleyI; 2026-10-05_Band1_Hadamard_13064_PaleyI;
2026-10-05_Band1_Hadamard_13044_PaleyI; 2026-10-05_Band1_Hadamard_13008_PaleyI;
2026-10-05_Band1_Hadamard_13004_PaleyI; 2026-10-05_Band1_Hadamard_12984_PaleyI;
2026-10-05_Band1_Hadamard_12980_PaleyI; 2026-10-05_Band1_Hadamard_12968_PaleyI;
2026-10-05_Band1_Hadamard_12960_PaleyI; 2026-10-05_Band1_Hadamard_12924_PaleyI;
2026-10-05_Band1_Hadamard_12920_PaleyI; 2026-10-05_Band1_Hadamard_12912_PaleyI;
2026-10-05_Band1_Hadamard_12908_PaleyI; 2026-10-04_Band1_Hadamard_12900_PaleyI;
2026-10-04_Band1_Hadamard_12824_PaleyI; 2026-10-04_Band1_Hadamard_12800_PaleyI;
2026-10-04_Band1_Hadamard_12792_PaleyI; 2026-10-04_Band1_Hadamard_12764_PaleyI;
2026-10-04_Band1_Hadamard_12744_PaleyI; 2026-10-04_Band1_Hadamard_12740_PaleyI;
2026-10-04_Band1_Hadamard_12704_PaleyI; 2026-10-04_Band1_Hadamard_12672_PaleyI;
2026-10-04_Band1_Hadamard_12660_PaleyI; 2026-10-04_Band1_Hadamard_12648_PaleyI;
2026-10-04_Band1_Hadamard_12620_PaleyI; 2026-10-04_Band1_Hadamard_12612_PaleyI;
2026-10-04_Band1_Hadamard_12584_PaleyI; 2026-10-04_Band1_Hadamard_12548_PaleyI;
2026-10-04_Band1_Hadamard_12540_PaleyI; 2026-10-03_Band1_Hadamard_12380_PaleyI;
2026-10-03_Band1_Hadamard_12348_PaleyI; 2026-10-03_Band1_Hadamard_12344_PaleyI;
2026-10-03_Band1_Hadamard_12324_PaleyI; 2026-10-03_Band1_Hadamard_12264_PaleyI;
2026-10-03_Band1_Hadamard_12252_PaleyI; 2026-10-03_Band1_Hadamard_12240_PaleyI;
2026-10-03_Band1_Hadamard_12228_PaleyI; 2026-10-03_Band1_Hadamard_12212_PaleyI;
2026-10-03_Band1_Hadamard_12204_PaleyI; 2026-10-03_Band1_Hadamard_12164_PaleyI;
2026-10-03_Band1_Hadamard_12144_PaleyI; 2026-10-03_Band1_Hadamard_12120_PaleyI;
2026-10-03_Band1_Hadamard_12108_PaleyI; 2026-10-03_Band1_Hadamard_12072_PaleyI;
2026-10-03_Band1_Hadamard_12044_PaleyI; 2026-10-03_Band1_Hadamard_12012_PaleyI;
2026-10-03_Band1_Hadamard_12008_PaleyI; 2026-10-03_Band1_Hadamard_11988_PaleyI;
2026-10-03_Band1_Hadamard_11972_PaleyI; 2026-10-03_Band1_Hadamard_11960_PaleyI;
2026-10-03_Band1_Hadamard_11940_PaleyI; 2026-10-03_Band1_Hadamard_11928_PaleyI;
2026-10-02_Band1_Hadamard_11924_PaleyI; 2026-10-02_Band1_Hadamard_11904_PaleyI;
2026-10-02_Band1_Hadamard_11888_PaleyI; 2026-10-02_Band1_Hadamard_11868_PaleyI;
2026-10-02_Band1_Hadamard_11864_PaleyI; 2026-10-02_Band1_Hadamard_11840_PaleyI;
2026-10-02_Band1_Hadamard_11832_PaleyI; 2026-10-02_Band1_Hadamard_11828_PaleyI;
2026-10-02_Band1_Hadamard_11808_PaleyI; 2026-10-02_Band1_Hadamard_11784_PaleyI;
2026-10-02_Band1_Hadamard_11780_PaleyI; 2026-10-02_Band1_Hadamard_11744_PaleyI;
2026-10-02_Band1_Hadamard_11732_PaleyI; 2026-10-02_Band1_Hadamard_11720_PaleyI;
2026-10-02_Band1_Hadamard_11700_PaleyI; 2026-09-30_Band1_Hadamard_11588_PaleyI;
2026-09-30_Band1_Hadamard_11580_PaleyI; 2026-09-30_Band1_Hadamard_11552_PaleyI;
2026-09-30_Band1_Hadamard_11528_PaleyI; 2026-09-30_Band1_Hadamard_11520_PaleyI;
2026-09-30_Band1_Hadamard_11504_PaleyI; 2026-09-30_Band1_Hadamard_11492_PaleyI;
2026-09-30_Band1_Hadamard_11484_PaleyI; 2026-09-30_Band1_Hadamard_11472_PaleyI;
2026-09-30_Band1_Hadamard_11468_PaleyI; 2026-09-30_Band1_Hadamard_11448_PaleyI;
2026-09-30_Band1_Hadamard_11444_PaleyI; 2026-09-30_Band1_Hadamard_11424_PaleyI;
2026-09-30_Band1_Hadamard_11412_PaleyI; 2026-09-30_Band1_Hadamard_11400_PaleyI;

2026-09-30_Band1_Hadamard_11384_PaleyI; 2026-09-30_Band1_Hadamard_11352_PaleyI;
2026-09-30_Band1_Hadamard_11312_PaleyI; 2026-09-30_Band1_Hadamard_11300_PaleyI;
2026-09-30_Band1_Hadamard_11288_PaleyI; 2026-09-30_Band1_Hadamard_11280_PaleyI;
2026-09-30_Band1_Hadamard_11252_PaleyI; 2026-09-29_Band1_Hadamard_11244_PaleyI;
2026-09-29_Band1_Hadamard_11240_PaleyI; 2026-09-29_Band1_Hadamard_11172_PaleyI;
2026-09-29_Band1_Hadamard_11160_PaleyI; 2026-09-29_Band1_Hadamard_11132_PaleyI;
2026-09-29_Band1_Hadamard_11120_PaleyI; 2026-09-29_Band1_Hadamard_11088_PaleyI;
2026-09-29_Band1_Hadamard_11084_PaleyI; 2026-09-29_Band1_Hadamard_11072_PaleyI;
2026-09-29_Band1_Hadamard_11060_PaleyI; 2026-09-29_Band1_Hadamard_11048_PaleyI;
2026-09-29_Band1_Hadamard_11028_PaleyI; 2026-09-29_Band1_Hadamard_11004_PaleyI;
2026-09-29_Band1_Hadamard_10988_PaleyI; 2026-09-29_Band1_Hadamard_10980_PaleyI;
2026-09-29_Band1_Hadamard_10940_PaleyI; 2026-09-29_Band1_Hadamard_10892_PaleyI;
2026-09-29_Band1_Hadamard_10884_PaleyI; 2026-09-29_Band1_Hadamard_10860_PaleyI;
2026-09-29_Band1_Hadamard_10848_PaleyI; 2026-09-29_Band1_Hadamard_10832_PaleyI;
2026-09-29_Band1_Hadamard_10800_PaleyI; 2026-09-29_Band1_Hadamard_10772_PaleyI;
2026-09-29_Band1_Hadamard_10740_PaleyI; 2026-09-29_Band1_Hadamard_10724_PaleyI;
2026-09-29_Band1_Hadamard_10712_PaleyI; 2026-09-28_Band1_Hadamard_10692_PaleyI;
2026-09-28_Band1_Hadamard_10688_PaleyI; 2026-09-28_Band1_Hadamard_10668_PaleyI;
2026-09-28_Band1_Hadamard_10652_PaleyI; 2026-09-28_Band1_Hadamard_10640_PaleyI;
2026-09-28_Band1_Hadamard_10632_PaleyI; 2026-09-28_Band1_Hadamard_10628_PaleyI;
2026-09-28_Band1_Hadamard_10608_PaleyI; 2026-09-28_Band1_Hadamard_10568_PaleyI;
2026-09-28_Band1_Hadamard_10560_PaleyI; 2026-09-28_Band1_Hadamard_10532_PaleyI;
2026-09-28_Band1_Hadamard_10500_PaleyI; 2026-09-28_Band1_Hadamard_10488_PaleyI;
2026-09-28_Band1_Hadamard_10464_PaleyI; 2026-09-28_Band1_Hadamard_10460_PaleyI;
2026-09-28_Band1_Hadamard_10428_PaleyI; 2026-09-28_Band1_Hadamard_10400_PaleyI;
2026-09-24_Band1_Hadamard_10392_PaleyI; 2026-09-24_Band1_Hadamard_10344_PaleyI;
2026-09-24_Band1_Hadamard_10332_PaleyI; 2026-09-24_Band1_Hadamard_10304_PaleyI;
2026-09-24_Band1_Hadamard_10272_PaleyI; 2026-09-23_Band1_Hadamard_10268_PaleyI;
2026-09-23_Band1_Hadamard_10224_PaleyI; 2026-09-23_Band1_Hadamard_10212_PaleyI;
2026-09-23_Band1_Hadamard_10164_PaleyI; 2026-09-23_Band1_Hadamard_10160_PaleyI;
2026-09-23_Band1_Hadamard_10152_PaleyI; 2026-09-23_Band1_Hadamard_10140_PaleyI;
2026-09-23_Band1_Hadamard_10112_PaleyI; 2026-09-23_Band1_Hadamard_10104_PaleyI;
2026-09-23_Band1_Hadamard_10100_PaleyI; 2026-09-23_Band1_Hadamard_10068_PaleyI.

Lab. `/workspace/experiments/band1/hadamard_13968_paleyI/`.

Cutoff. 2026-10-08T02:14:42Z (~10:14PM ET).

Honesty stamp (non-negotiable).

This memo claims **finite construction of order 13968 only (verified max|HHˆT-nI|=0), via classical Paley-I q=13967 (prime ===3 mod 4). 13968 IN SPK via Paley-I.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 13968: Paley-I **PASS** ($q = 13967$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL/SKIP** (q=6983 prime but ===3 mod 4 needs ===1); Kronecker **SKIP deferred note-only** (note $13968 = 4 \cdot 3492$; $3492 \equiv 0 \pmod 4$

order-admissible; factors $2^2 \cdot 3^2 \cdot 97$; not_in_seeds / no H_3492; H_3492 classically via Paley-I q=3491 prime ===3 mod 4 (direct) ALSO ELIGIBLE – deferred note-only, NOT constructed, not on box). **13968** folds into SPK **only** via classical Paley-I (this run). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **13524+13500+13488+13464+13452+13412+13400+13368+13340+13332+13328+13292+13268+132** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

ewpage

Abstract

Finite construction of order **13968**.

- H_{13968} = Paley-I ($q = 13967$) – classical SPK
- Paley-II for 13968: **FAIL/SKIP** ($q = 6983$ prime but $\equiv 3 \pmod 4$ needs $\equiv 1$)
- Kronecker for 13968: **SKIP deferred note-only** ($4 \cdot 3492$; $3492 \equiv 0 \pmod 4$ order-admissible; factors $2^2 \cdot 3^2 \cdot 97$; not_in_seeds / no H_3492; H_3492 classically via Paley-I q=3491 prime ===3 mod 4 (direct) ALSO ELIGIBLE – deferred note-only, NOT constructed, not on box)
- Skip: Miyamoto digression this wave

SPK: IN SPK via Paley-I.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_13968.py`

Results. overall_pass=True; units 632/632; row blocks 110/110 (block size 128); wall 2573.2223s; constructed max|HHˆT-nI|=0.

Cutoff: 2026-10-08T02:14:42Z (~10:14PM ET)

Arithmetic

- Target $n = 13968 = q + 1$ with $q = 13967$ prime $\equiv 3 \pmod 4$ — Paley-I eligible.
- Paley-II: $q = 6983$ prime but $\equiv 3 \pmod 4$ needs $\equiv 1$ — FAIL/SKIP.
- Kronecker: $13968 = 4 \cdot 3492$; $3492 \equiv 0 \pmod 4$ order-admissible; factors $2^2 \cdot 3^2 \cdot 97$; not_in_seeds / no H_3492; H_3492 classically via Paley-I q=3491 prime ===3 mod 4 (direct) ALSO ELIGIBLE – deferred note-only, NOT constructed, not on box — SKIP deferred note-only.
- Miyamoto digression skipped this wave (prefer classical SPK).

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_13968 Paley-I	Paley-I q=13967	0	2535.4177
H_13968 Paley-II	FAIL/SKIP (q=6983 prime but ===3 mod 4 needs ===1)	–	–
H_13968 Kronecker	SKIP deferred note-only (4*3492; 3492===0 mod 4 order-admissible; factors 2ˆ23ˆ2*97; not_in_seeds / no H_3492; H_3492 classically via Paley-I q=3491 prime ===3 mod 4 (direct) ALSO ELIGIBLE – deferred note-only, NOT constructed, not on box)	–	–

SPK closure note

Seeds include prior SPK through **11528** lineage (**11528**; **11520**; **11504**; **11492**; **11484**; **11472**; **11468**; **11448**; **11444**; **11424**; **11412**; **11400**; **11384**; **11352**; **11312**; **11300**; **11288**; **11280**; **11252**; **11244**; **11240**; **11172**; **11160**; **11132**; **11120**; **11088**; **11084**; **11072**; **11060**; **11048**; **11028**; **11004**; **10988**; **10980**; **10940**; **10904**; **10892**; **10884**; **10860**; **10848**; **10832**; **10800**; **10772**; **10740**; **10724**; **10712**; **10692**; **10688**; **10668**; **10652**; **10640**; **10632**; **10628**; **10608**; **10568**; **10560**; **10532**; **10500**; **10488**; **10464**; **10460**; **10428**; **10400**; **10392**; **10344**; **10332**; **10304**; **10272**; **10268**; **10260**; **10248**; **10244**; **10224**; **10212**; **10164**; **10160**; **10152**; **10140**; **10112**; **10104**; **10100**; **10068**; **10040**; **10008**; **9968**; **9932**; **9924**; **9908**; **9888**; **9884**; **9872**; **9860**; **9852**; **9840**; **9812**; **9804**; **9792**; **9788**; **9768**; **9744**; **9740**; **9720**; **9680**; **9644**; **9632**; **9624**; **9620**; **9588**; **9552**; **9548**; **9540**; **9512**; **9492**; **9480**; **9468**; **9420**; **9404**; **9392**; **9372**; **9344**; **9324**; **9320**; **9312**; **9284**; **9240**; **9228**; **9204**; **9200**; **9188**; **9152**; **9104**; **9128**; **9124**; **9092**;

2220+8884; **2236**; **2160**; **2140+8564**; . . . ; **NOT** 8948/8644/8516/. . .) + verified **10464** (+ prior **10392+10344+10332+10304+10272+10268+10224+10212+10164+10160+10152+10140+10112+101**
Next constructive gap **2052** (OPEN).

Quantity	Value
12488 in SPK	**True** (Paley-I q=13967)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

Next optional

q2=6961->n=13924 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6961 prime ===1 mod 4; Paley-I q=13923=3^2*71317 *NOT prime -> FAIL; Kronecker 13924=4*3481; 3481=59^2 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3481); 13924=2^2*59^2 *===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=6977->n=13956 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6977 prime ===1 mod 4; Paley-I q=13955=5*2791 NOT prime -> FAIL; Kronecker 13956=4*3489; 3489=3*1163 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3489); 13956=2^2*3*1163 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=6997->n=13996 Paley-II DIRECT (FIRST n===0 mod 4 after 13968 with any classical Paley route; 13972..13992 have none; q2=6997 prime ===1 mod 4; Paley-I q=13995=3^2*5*311 NOT prime -> FAIL; Kronecker 13996=4*3499; 3499 odd prime (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3499); 13996=2^2*3499 ===12 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | next Paley-I DIRECT: q=13999->n=14000 (prime ===3 mod 4; 13997 prime ===1 mod 4 skipped; no prime ===3 mod 4 in 13968..13998; Paley-II q=6999=3*2333 *NOT prime; ===3 mod 4 needs ===1 -> FAIL/SKIP; Kronecker 14000=4*3500; 3500=2^2*5^3* ===0 mod 4 order-admissible, NOT in ONBOX_SEEDS; H_3500 classically via Paley-I q=3499 prime ===3 mod 4 (direct) - ===0 mod 16 = 13968/13568-like Kronecker shape, Paley-II composite ===3 as 13568) (or gap 2052 / Track B)